IKT215 Exercise 4: Parametric Classification

February 12th, 2026

Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails (manuel.s.mathew@uia.no).

Question 1: Maximum Likelihood Estimation (MLE)

Problem Statement:

Given a dataset $X = \{x_1, x_2, \dots x_n\}$ is drawn from a normal distribution $\mathcal{N}(\mu, \sigma^2)$. The probability density function (PDF) for a single observation x_i is:

$$f(x_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Part 1: Theoretical Derivation

Derive the maximum likelihood estimates for:

1. Population mean, μ, and
2. Population variance, σ^2.

Show all steps of the derivation.

Part 2: Implementation and Verification

Implement the formulas derived in Python and verify them using:

1. A synthetic dataset with known parameters
2. Compare with theoretical values
3. Do not use `numpy` for an additional challenge (Optional)

Part 3: Analysis Questions

1. How do the MLE estimates compare to the true parameters?
2. What happens to the accuracy of the estimates as you increase the sample size?
3. Why might the MLE estimate for variance be biased?

Question 2: Maximum Likelihood Estimation (MLE)

Problem Statement:

Given a dataset of 100 independent coin flips where 62 outcomes are heads and 38 are tails, derive the maximum likelihood estimate (MLE) for the probability p of observing heads using the Bernoulli distribution. Bernoulli probability mass function is given by

$$P(X = x) = p^x(1-p)^{1-x} \text{ for } \} \; x \in \{0,1\}.$$

Question 3: Linear Discriminant Analysis

Problem Statement

Generate a synthetic dataset with 2 classes following multivariate normal distributions with identical covariance matrices but different means. Implement LDA classification with the following requirements:

1. Split data into 70% training and 30% test sets
2. Train LDA classifier on training data
3. Report test accuracy and plot decision boundary
4. Visualize class distributions with decision boundary (Recommended)

Question 4: Quadratic Discriminant Analysis

Problem Statement

Generate a synthetic dataset with 2 classes having different covariance matrices. Implement QDA classification with:

1. 60% training and 40% test split
2. QDA model training
3. Test set evaluation with confusion matrix
4. Non-linear decision boundary visualization (Recommended)
5. Apply the LDA on this data and observe the change in accuracy